EEI6373 Performance Modelling

Mini Project

Bachelor of Software Engineering

Department of Electrical and Computer Engineering

The Open University of Sri Lanka

Name: P.G.P Perera

S-Number: S92065006

RegNo: 321425006

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Performance Modelling of a Smart Traffic Light Control System

Abstract

Traffic congestion is becoming a significant issue in urban areas today, as there is an increase in the number of vehicles on the road and traffic management practices remain less efficient. The traditional traffic light systems, which depend on fixed intervals, cannot change according to the traffic situations on roads. As a result, drivers have to wait more time, and traffic jams become longer thus causing inefficiency in the functioning of the transport system. The research reported herein is based on the study of the Smart Traffic Light System in terms of its performance that employs an embedded control and uses both the sensors and smart algorithms for adjusting the traffic light intervals.

The purpose of the research is to assess the operation of such sort of the system from the perspective of the parameters of its performance efficiency and effectiveness when experiencing a different vehicular load. The methodology employed in the project is based on computer simulations that helped to create models of different traffic situations. Theoretical methodologies of queueing theory as well as performance analysis and evaluation techniques were applied in the research process.

1. System Description

A Smart Traffic Light Control System is an advanced embedded system that is implemented to enhance the traffic at the junctions of roads. Instead of following a fixed schedule, this system evaluates the traffic situation and adapts the timings of signals according to the present traffic condition.

The components of the system include the following:

- Vehicle detection sensors/cameras
- Embedded traffic controller
- Communication network
- Traffic signal control units
- Central monitoring server

The vehicle detection sensors give information about the number of vehicles coming to each road. This information is sent to the embedded controller that processes the data and calculates the optimal duration for green light for vehicles approaching each road. The purpose of the controller is to minimize the wait time of cars and maximize the number of vehicles that can pass through the intersection.

The system might have performance issues when there is an increasing traffic volume. High traffic density leads to longer processing times, higher controller load, delays in communication, and longer vehicle queues. Due to that, it is important to conduct performance modeling in order to evaluate system performance and detect possible points of performance deterioration. [1]

2. Performance Goals

The main performance objectives of this project are:

Objectives	Description
Minimize Response Time	The traffic controller must process sensor information and update signal timings with minimum delay to support real-time decision making.
Maximize Throughput	The system should maximize the number of vehicles passing through the intersection within a given time period.
Reduce Vehicle Waiting Time	The system should minimize the average waiting time experienced by vehicles at the intersection.
Identify Performance Bottlenecks	The model should identify limitations caused by sensor processing, controller workload, communication delays, or excessive traffic volume.
Optimize Resource Utilization	The embedded controller should operate efficiently without excessive CPU utilization or memory consumption.
Evaluate Scalability	The system performance should be evaluated when traffic volume increases to determine the maximum load that can be handled efficiently.

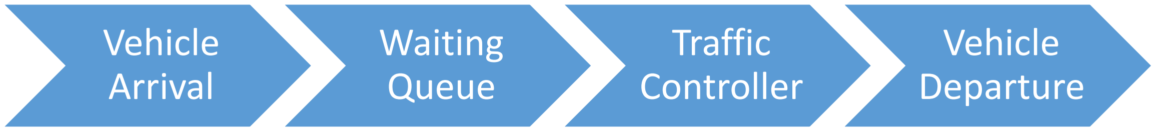
3. Modeling Approach and Assumptions

A combination of queueing modelling and simulation-based performance analysis is used for this project.

3.1 Queueing Model

The intersection is modelled as a queueing system where vehicles arrive at the intersection and wait until the traffic signal allows them to proceed.

The system can be represented using a queueing model. [2]



A queueing approach is suitable because vehicles arrive randomly and require service time based on traffic signal availability.

The main parameters considered are:

- Arrival rate (λ): Number of vehicles arriving per minute
- Service rate (μ): Number of vehicles that can pass through the intersection per minute
- Queue length: Number of waiting vehicles
- Waiting time: Time spent before vehicles pass

3.2 Simulation Model

A simulated dataset is generated to represent different traffic conditions ranging from low traffic to heavy congestion. [2]

The following assumptions are made:

- Vehicle arrivals follow a variable traffic pattern.
- Sensor detection is assumed to be accurate.
- Communication delay between sensors and controller is constant.
- The controller has limited processing capacity.
- Environmental factors such as weather conditions are not considered.
- Traffic rules and pedestrian crossings are not included in the current model.

4. Data Description and Methodology

The dataset represents the performance of the Smart Traffic Light Control System under different traffic volumes.

The following parameters are measured:

Traffic Volume (Vehicles/min)	Response Time (ms)	Waiting Time (s)	Queue Length	CPU Utilization (%)	Throughput (Vehicles/min)
50	60	8	4	20	48
100	75	12	8	32	96
150	95	18	15	48	142
200	125	27	24	63	185
250	165	38	35	78	225
300	220	55	48	91	255

The dataset is analyzed using performance metrics to understand how increasing traffic volume affects system behavior. [3]

Graphs are generated to visualize:

- Traffic volume vs response time
- Traffic volume vs queue length
- Traffic volume vs CPU utilization
- Traffic volume vs throughput

5. Comprehensive Examination and Results

5.1 Evaluation of Response Time

The evaluation reveals that with rising traffic intensity, the response time of the controller tends to increase. Specifically, the response time is about 60 milliseconds at low traffic (50 vehicles/min) but will rise to 220 milliseconds at 300 vehicles/min. [4] [5]

This further suggests that increases in traffic intensity lead to an increase in processing requirements, thereby creating a delay in decision-making.

5.2 Evaluation of Queue Length

The length of the queue grows as traffic density increases. Thus, the length of the queue increase from only 4 vehicles at 50 vehicles/min to about 48 vehicles at 300 vehicles/min.

This means that when vehicle arrival rate approaches maximum service capacity, traffic congestion grows significantly.

5.3 Examination of CPU Utilization

CPU utilization rises from 20% to 91% as traffic intensity rises. This proves that the embedded controller will have to handle higher computation loads during peak traffic periods.

Once the CPU utilization exceeds 80%, the system may not function properly.

5.4 Throughput Evaluation

The throughput increases in magnitude initially as traffic volume rises. However, once higher traffic levels have been reached, the further throughput gains will be limited since the intersection will reach its processing

This indicates a scalability limitation in the current configuration.

6. Visualizations

The following visualizations are included in the report:

Figure 1: Smart Traffic Light System Architecture

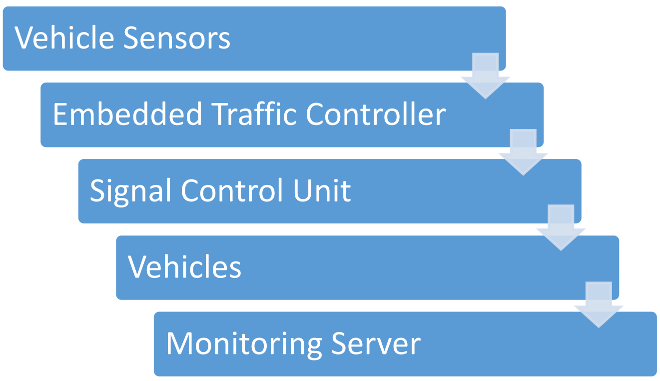


Figure 2: Traffic Volume vs Response Time

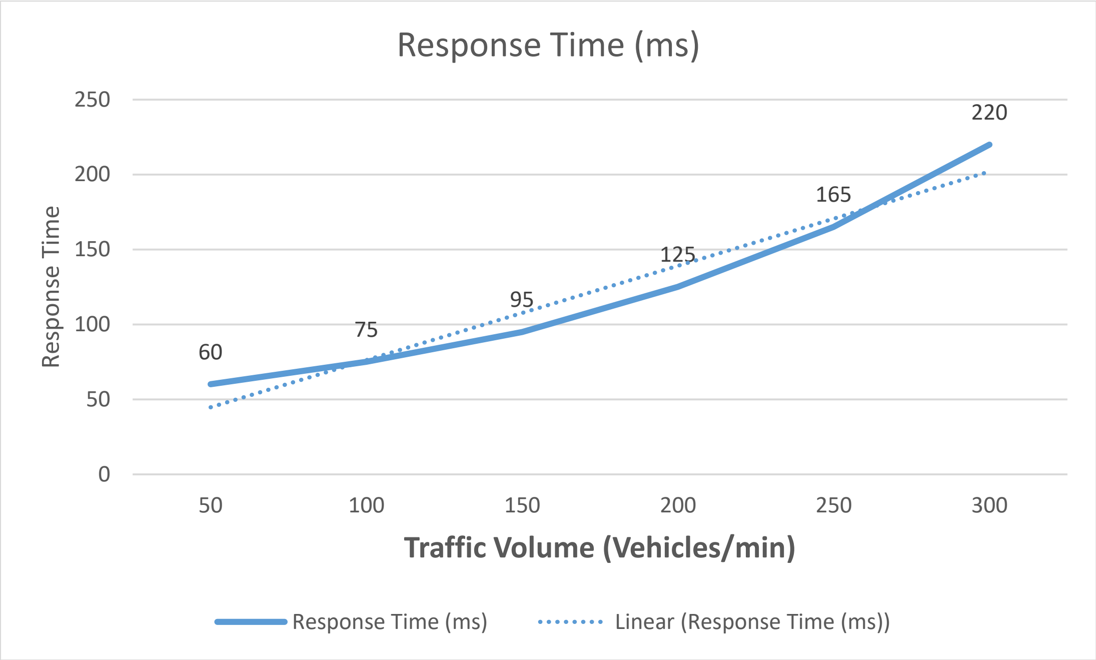


Figure 3: Traffic Volume vs Queue Length

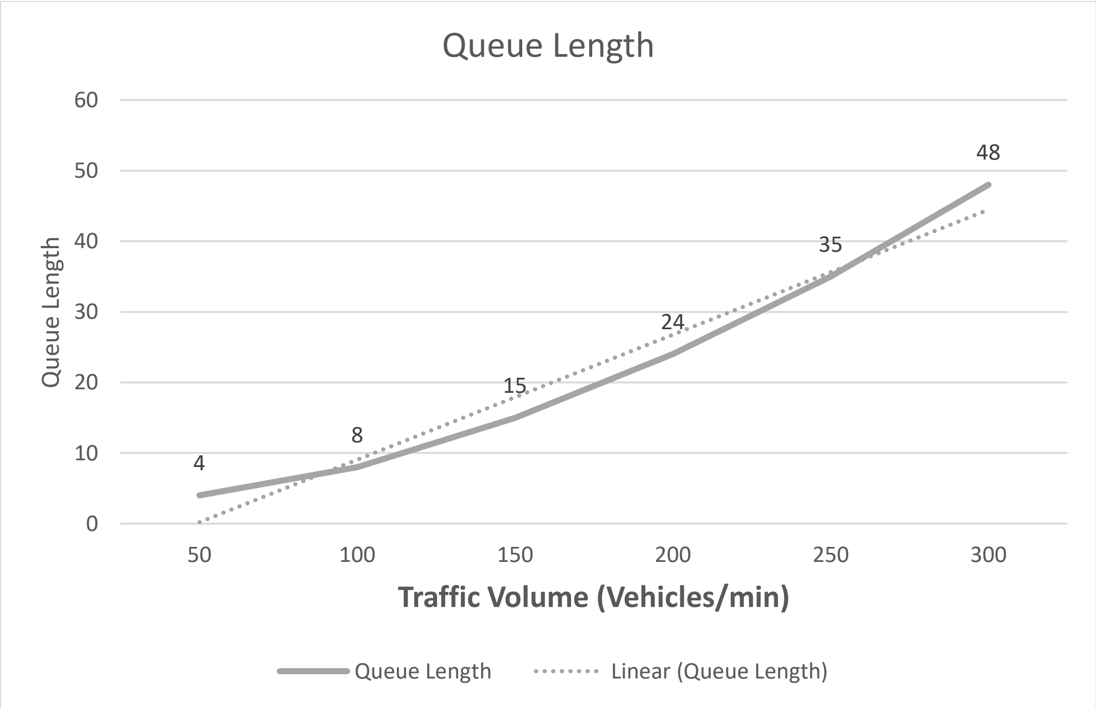


Figure 4: Traffic Volume vs CPU Utilization

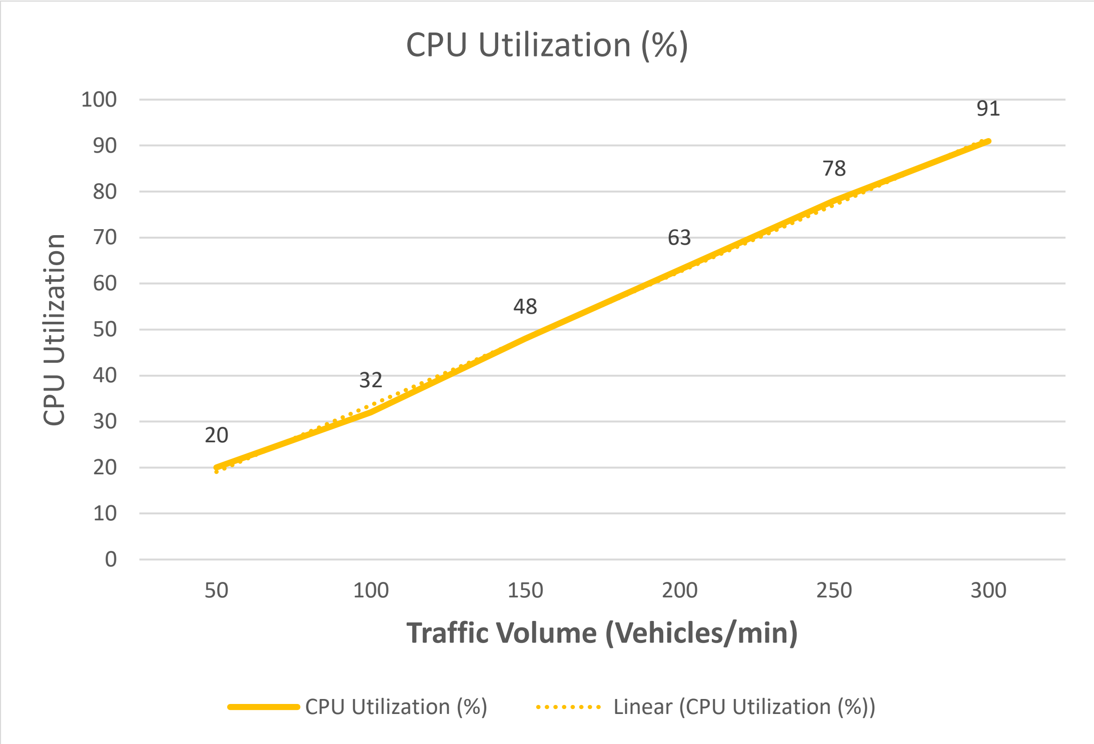
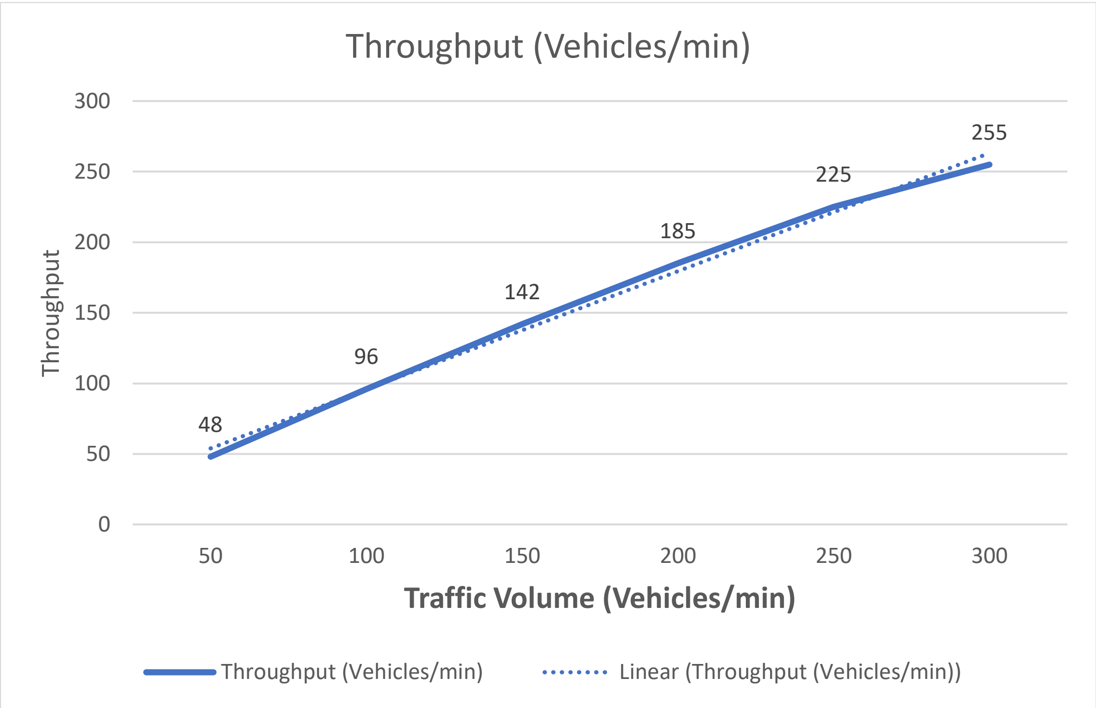


Figure 5: Traffic Volume vs Throughput



7. Performance Improvement Recommendations

Based on the analysis, several improvements can be proposed:

- Implement edge computing to reduce communication delays.
- Upgrade embedded controller processing capability.
- Use machine learning algorithms for traffic prediction.
- Deploy additional sensors for better traffic estimation.
- Implement adaptive signal timing algorithms.
- Use distributed traffic management for multiple intersections.

8. Limitations

The current model has several limitations:

- The dataset is simulated rather than collected from a real intersection.
- Weather and road conditions are not considered.
- Pedestrian movement is not included.
- Communication failures are not modelled.
- The model represents a single intersection only.

9. Future Extensions

Future improvements can include:

- Integration with real IoT traffic sensors.
- Development of a multi-intersection traffic simulation model.
- Use of machine learning for predictive traffic control.
- Real-time traffic data collection from smart cities.
- Integration with emergency vehicle priority systems.

10. Conclusion

The study investigates the efficiency of Smart Traffic Light Control System through the use of performance modelling techniques. The study looked at the system under various traffic loads, evaluating the performance based on response time, throughput, queue length, and CPU utilization.

The results indicate that the increase in traffic volume leads to the increase in response times, queue lengths, and resource utilization of the controller. Moreover, it helps to pinpoint the performance limitations and suggest improvements.

Performance modeling is an effective method for analyzing the behavior of intelligent transport systems and helps in making better decisions prior to using the systems in practice.

*** End ***

References

[1] B. G. R and K. NAGAMANI, “IMPLEMENTATION OF INTELLIGENT TRAFFIC MANAGEMENT SYSTEM USING IOT,” *INTERNATIONAL JOURNAL OF ELECTRICAL ENGINEERING AND TECHNOLOGY*, vol. 11, no. 5, July 2020, doi: 10.34218/ijeet.11.5.2020.003.

[2] L. Kleinrock, *Queueing Systems, Volume 1: Theory*, New York: John Wiley & Sons, 1975.

[3] R. Jain, *The Art of Computer Systems Performance Analysis: Techniques for Experimental Design, Measurement, Simulation, and Modeling*, New York: Wiley, 1991.

[4] “Google Books,” *Google.lk*, 2026. https://www.google.lk/books/edition/Advances_in_Artificial_Transportation_Sy/OxykAwAAQBAJ?hl=en&gbpv=1&dq=M.+Abadi+and+M.+K.+Hassan (accessed Aug. 08, 2026).

[5] “Google Books,” *Google.lk*, 2022. https://www.google.lk/books/edition/Smart_Transportation_Systems_2022/p8VvEAAAQBAJ?hl=en&gbpv=1&dq=M.+Abadi+and+M.+K.+Hassan (accessed Aug. 08, 2026).

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